DE–3593 **11**

DISTANCE EDUCATION

B.C.A. DEGREE EXAMINATION, MAY 2012.

ELECTRONIC DEVICES AND DIGITAL CIRCUITS

(2003 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

(5 × 20 = 100)

1. (a) Explain in detail about BCD codes, Gray code, ASCII code with suitable example. (10)

   (b) (i) Convert $(1244)_{10}$ to $(?)_2$ (3)

   (ii) Convert $(11101101101011)_2$ to $( )_{16}$ (3)

   (iii) Convert $(248)_{16}$ to $( )_{10}$. (4)

2. (a) Explain how the transistor acts as a switch. (10)

   (b) Draw the logic symbol for AND, OR , NOT, and X-OR and give its truth table. (10)

3. (a) (i) Add $(1111)_2$ and $(0101)_2$

   (ii) Find 2's complement of $(1010101)_2$

   (iii) Write the signed binary number for $(+8)_{10}$

   (iv) Subtract $(0110)_2$ from $(1011)_2$. $(4 \times 2\frac{1}{2} = 10)$

   (b) Prove the following using De Morgan's theorem $AB + CD = \overline{\overline{AB}.\overline{CD}}$ (10)

4. (a) Draw the half subtractor logic circuit and explain it with truth table. (10)

(b) Draw the three variable minterm karnaugh map and explain how it is used for simplification. (10)

5. (a) (i) Draw R-S filp flop and give its truth table. (5)

(ii) Draw the D filp-flop and give the truth table. (5)

(b) Explain the function of 3-bit binary counter with truth table. (10)

6. (a) Draw the full adder logic circuit and explain it with truth table. (10)

(b) Briefly explain about Quine Mc kluskey tabulation method. (10)

7. (a) Explain working function and characteristics of a PN junction diode. (10)

(b) Explain the function of MOSFET. (10)

8. (a) Explain the function of a photodiode. (10)

(b) With a neat diagram explain the function of silicon control rectifier. (10)

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DE–3594 12

DISTANCE EDUCATION

B.C.A. DEGREE EXAMINATION, MAY 2012.

PRINCIPLES OF INFORMATION TECHNOLOGY

(2003 Onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

(5 × 20 = 100)

1. (a) Explain the six elements of a computer and communications system.

   (b) Describe the development of computer technology.

2. (a) Discuss about the kinds of software.

   (b) Describe the graphics presentation software.

3. (a) How do you use the communications and connectivity practically? Explain.

   (b) Discuss and mention the factors affecting the data transmission.

4. (a) Discuss about the video conferencing.

   (b) Compare and contrast the internet and intranet.

5. (a) Describe the fundamentals of storage and its uses.

   (b) Explain the features of a DBMS.

6. (a) How can you organise the data in a secondary storage system? Discuss.

(b) With a suitable example, explain the compression and decompression.

7. (a) Discuss in detail about the MIS.

(b) Discuss the five steps in programming.

8. (a) Discuss about the internet programming.

(b) Discuss about the object oriented programming in detail.

DE–3595 | **13**

DISTANCE EDUCATION

B.C.A. DEGREE EXAMINATION, MAY 2012.

PRINCIPLES OF OPERATING SYSTEM

(2003 onwards)

Time : Three hours | Maximum : 100 marks

Answer any FIVE questions.

Each question carries equal marks.

$(5 \times 20 = 100)$

1. (a) Why do you called an operating system as a resource manager? Discuss.

   (b) Explain the history of an operating system.

2. (a) Describe in detail about the semaphore.

   (b) With suitable example, explain any two process scheduling algorithms.

3. (a) Explain Derive-independent I/O software.

   (b) How can you detect and recovery the deadlochs? Explain with suitable example.

4. (a) Explain multiprogramming without swapping or paging.

   (b) Discuss briefly about the linked list and buddy systems.

5. (a) How can you analyse the swapping system? Discuss with algorithm.

   (b) Describe the page replacement algorithms.

6. (a) Explain about the disk space management.

(b) Explain about the file servers.

7. (a) Explain multiprogramming with variable partition.

(b) Explain memory management with bit-maps.

8. (a) Discuss about the derive drivers with examples.

(b) Explain about the interrupt handler.

DE–3596 **14**

DISTANCE EDUCATION

B.C.A. DEGREE EXAMINATION, MAY 2012.

C PROGRAMMING

(2003 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

(5×20 = 100)

1. (a) Explain the top down approach in program development. (10)

   (b) Explain any five data types in C with example. (10)

2. (a) (i) Write the character set of C. (5)

   (ii) Illustrate the symbolic constant with example program. (5)

   (b) Explain the control statements

   (i) for (5)

   (ii) do …… while with example (5)

3. (a) Write a C program to find the roots of a given quadratic equation also classify its roots. (10)

   (b) Explain the concept of arrays and how it can be manipulated in C language. (10)

4. (a) Write a C program to sort the given N names in alphabetical order. (10)

   (b) Explain the role of functions in C with example. (10)

5. (a) Discuss on the recurstion function in C with example program. (10)

(b) Explain the concept of passing pointes as arguments in functions with example. (10)

6. (a) Explain the storage classes

(i) auto (5)

(ii) register with example (5)

(b) Create an employee structure with the fields empno, empname, basicpay for ten employees. Calculate the netpay with the allowances DA, HRA and deductions PF, tax of each employee and display them. (10)

7. (a) Explain the concept of streams with example. (10)

(b) Describe the various operations on files. (10)

8. (a) Explain the macro substitutions with example. (10)

(b) Write a file program to read the students information from a data file and display them. (10)

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DE–3597 **15**

DISTANCE EDUCATION

B.C.A. DEGREE EXAMINATION, MAY 2012.

ACCOUNTING FUNDAMENTALS

(2003 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

(5 × 20 = 100)

1. Differentiate between financial accounting and cost accounting. (20)

2. Explain the conventions of accounting and describe the historical functions of accounting. (10 + 10)

3. What do you mean by double entry system of book-keeping? What is classification of accounts? Explain the rules for making entries under double entry system. (6 + 6 + 8)

4. Describe the limitations of ratios. (20)

5. Enter the following transactions in three column cash book. (20)

2007 Jan		Rs.
1	Cash in hand	410
	Balance at bank	8,920

2007 Jan		Rs.
2	Cash sales	4,500
3	Paid into bank	4,000
5	Purchased stationery	100
8	Paid Mahendran by cheque	280
	Discount received	20
12	Gave a cheque for cash purchases	1,500
15	Drew for personal use	500
18	Received from Sankar, a cheque for Rs. 1,970 in full settlement of Rs. 2,000	
20	Drew from Bank	1,000
21	Paid wages	800
31	Bank charges as per pass book	10

6. From the following Trial balance extracted from the books of Kamal prepare trading, profit and loss account and balance sheet for the year ended 31.12.2006. (20)

Debit Balances	Rs.	Credit Balances	Rs.
Cash at Bank	2,610	Creditors	4,700
Book debts	11,070	Discounts	150
Salaries	4,950	Outstanding expenses	400
Carriage in	1,450	Return outwards	2,520
Carriage out	1,590	Sales	80,410
Bad debts	1310	Capital	40,000
Office expenses	5,100		
Purchases	67,350		
Return inwards	1,590		

Debit Balances	Rs.	Credit Balances	Rs.
Furniture	1,500		
Stock	14,360		
Insurance	3,300		
Depreciation on property	1,200		
Freehold property	10,800		
	1,28,180		1,28,180

Closing stock was valued at Rs. 5,340.

7. Ramesh keeps his books on single entry basis. Prepare a statement of affairs as on 31.10.2007 and a statement of profit or loss for the period ending 31.10.07. (20)

Assets and liabilities	1.11.06 Rs.	31.10.07 Rs.
Bank balance	560	350
Cash on hand	10	50
Debtors	4,500	3,600
Stock	2,700	2,900
Plant	4,000	4,000
Furniture	1,000	1,000

Ramesh had withdrawn Rs. 2,000 during the year ended 31.10.07 and had introduced fresh capital of Rs. 4,200 on 1.7.07. A provision of 5% on debtors in necessary. Write off depreciation on plant at 10% and furniture at 15%.

8. From the following details find out :

(a) Current assets. (5)

(b) Current liabilities (5)

(c) Liquid assets. (5)

(d) Stock. (5)

Current ratio	:	2.8
Liquid ratio	:	1.5
Working capital	:	Rs. 1,62,000.

DE–4352 **16**

DISTANCE EDUCATION

B.C.A. DEGREE EXAMINATION, MAY 2012.

LAB – I : C PROGRAMMING

(2003 onwards)

Time : Three hours Maximum : 100 marks

Examiner has to select and give ONE question to each candidate by lot system.

Each subdivision carries 50 marks.

1. (a) There are 10 students in a class. Their names and marks in five different subjects are given. If a student takes more than 40 marks in each subject then he is declared "PASS". Otherwise "FAIL". Write a C program to do the above.

   (b) Write a C program to copy the content of one file to another.

---------- Cut here ----------

2. (a) Write a C program to accept two strings. Compare whether the strings are equal or not and then concatenate the two strings and find the length of concatenated string.

   (b) Write a C program to create an ADDRESS file.

3. (a) Write a C program to sort the given set of numbers in ascending order or descending order on request basis.

(b) Write a C program to accept and display your ADDRESS using STRUCTURE.

---------- Cut here ----------

4. (a) Write a C program to print the all ASCII Characters.

(b) Write a C program to create a text file and display the contents of the text file.

---------- Cut here ----------

5. (a) Write a C program to find the word is palindrome or not.

(b) Write a C program to accept and display your Date of Birth using STRUCTURE.

---------- Cut here ----------

6. (a) Write a C program to find the product of two matrices.

(b) Write a C program to read the records in an ADDRESS file.

7. (a) A man is paid at the hourly rate of Rs.15/– per hour for the first 45 hours worked. Thereafter, overtime is paid at 1.5 times the hourly rate for the next 25 hours and two times the hourly rate for further worked per week. Calculate and print his weekly wage.

(b) Write a C program to arrange the names in a class in ascending order and assign roll number to each candidate.

---------- Cut here ----------

8. (a) Write a C program that receives the data such as age and name of person to check the eligibility for voting. Take the condition that if a person is more than 18 years, he is eligible to vote. Else display the number of years, he has to wait for voting.

(b) Write a C program to display the address and the contents of pointer variable.

DE–3598 **21**

DISTANCE EDUCATION

B.C.A./B.C.A. (Lateral) DEGREE EXAMINATION,
MAY 2012.

MICROPROCESSORS AND PERSONAL COMPUTERS

(2003 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks. (5 × 20 = 100)

1. (a) With a suitable functional diagram. Explain the address bus and control bus. (10)

   (b) Write a detailed note on "Evolution of Microprocessor". (10)

2. (a) Draw the functional block diagram of 8085. Explain its architecture. (10)

   (b) What is an interrupt? Explain about the Software and Hardware interrupts. (10)

3. (a) List out the addressing modes that are found in 8085. Explain each with an example. (10)

   (b) Explain what operation will take place when the following instructions are executed in 8085

      (i) LXI r_p, data

      (ii) LDA addr

      (iii) LHLD addr

      (iv) STA addr

      (v) SHLD addr. (5 × 2 = 10)

4. (a) Write an AL program to add two hex numbers CB and E9. After the execution of the program, give the status of various flag. (10)

(b) Write an AL program to multiply two 8 bit numbers stored in memory location 8C 00 and 8C 01 and place the result in memory location 8C 02. (10)

5. (a) Give a short note about the memory Hierarchy. (5)

(b) Discuss in detail about the RAM and various types of ROM. (15)

6. (a) Explain what is memory mapped I/O scheme and I/O mapped I/O scheme. (10)

(b) Explain the working of Direct Memory Access data transfer. (10)

7. (a) Discuss in detail about the internal organization of 80486. (10)

(b) Give a brief note about the following

(i) SCSI interface (5)

(ii) SUGA Video Card. (5)

8. (a) Discuss about the passive preventive maintenance procedure in detail. (12)

(b) Give a brief note about the data backup system. (8)

DE–3599 **22**

DISTANCE EDUCATION

B.C.A./B.C.A.(Lateral) DEGREE EXAMINATION, MAY 2012.

OFFICE AUTOMATION

(2003 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks ($5 \times 20 = 100$)

1. (a) List and Explain the purpose of menu bar and tool bars. (10)

   (b) What is a document? Write about renaming, copying and moving a document. (10)

2. (a) How can you edit, select, copy and format a text in word document? Explain. (10)

   (b) Explain about inserting, moving and resizing pictures in word document. (10)

3. (a) What is a worksheet? Explain about adding and removing rows and columns in worksheet. (10)

   (b) How can you use chart wizard? Describe about creating, enhancing and printing a chart. (10)

4. (a) Explain about create and saving a table in access database. (10)

   (b) How can you create and print reports in access? Explain. (10)

5. (a) How can you integrate office applications in powerpoint presentation? Explain. (10)

(b) Explain about creating and printing binders. (10)

6. (a) Write about moving, correcting and inserting a text in word document. (10)

(b) What is a spellcheck? How will you carryout the spellcheck in word document? (10)

7. (a) Explain about creating and copying formula in worksheet with example. (10)

(b) How can you improve the appearance of worksheet? Explain. (10)

8. (a) What is a record? Explain about various operations on records. (10)

(b) Explain in detail query and forms in access. (10)

DE–3600 **23**

DISTANCE EDUCATION

B.C.A./B.C.A./(Lateral) DEGREE EXAMINATION, MAY 2012.

SYSTEMS ANALYSIS AND DESIGN

(2003 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions. (5 × 20 = 100)

All questions carry equal marks.

1. Explain the different phases of system development life cycle. (20)

2. Explain the following fact finding techniques:

   (a) Interview

   (b) Questionnaire. (10+10)

3. Explain the following structured analysis tools.

   (a) Decision tree

   (b) Structured English. (10+10)

4. (a) Summarize the points to be considered for designing the printed output. (10)

   (b) Discuss the input validation process. (10)

5. Explain the following types of databases:

   (a) Hierarchical

   (b) Relational. (10+10)

6. (a) List out the different types of files and explain each one of them. (10)

(b) Draw the symbols of data flow diagram and state their meaning. Illustrate the usage of data flow diagram with suitable example. (10)

7. (a) Give an outline about Top down system design approach. (10)

(b) Write a note on special system tests. (10)

8. (a) Explain the tasks involved in the conversion process. (10)

(b) Mention the importance of system maintenance and discuss the jobs involved in system maintenance.(10)

DE–3601 **24**

DISTANCE EDUCATION

B.C.A./B.C.A. (Lateral) DEGREE EXAMINATION, MAY 2012.

OBJECT ORIENTED PROGRAMMING AND C++

(2003 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

(5 × 20 = 100)

1. What are the main conceptual differences between object oriented programming and the other programming techniques?

2. What is meant by function overloading? Write the rules associated with function over loading.

3. (a) Explain the operators in C++.

   (b) Write a note on static data and member function.

4. (a) What is an abstract class? Give an example.

   (b) What is this pointer? State its use with example.

5. Describe upon the inheritance methods and explain with an example.

6. What is a constructor? Explain any two types of constructor with example.

7. What is operator overloading? Give an example for unary operator overloading.

8. Write a C++ program to generate the following figure using loop construct :

2	4						
3	6	9					
4	8	12	16				
5	10	15	20	25			
6	12	18	24	30	36		
7	14	21	28	35	42	49	
8	16	24	32	40	48	56	64

DE-3602 | **25**

DISTANCE EDUCATION

B.C.A /B.C.A (Lateral) DEGREE EXAMINATION, MAY 2012.

COMPUTER GRAPHICS

(2003 onwards)

Time : Three hours | Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

(5 × 20 = 100)

1. (a) Explain video display generation. (10)

   (b) Explain hardcopy output devices. (10)

2. (a) Explain the simple DDA algorithm with diagram. (10)

   (b) Write Ellipse algorithm and explain. (10)

3. (a) What is transformation principles and explain all two dimensional transformations? (15)

   (b) Write short notes on 2-D composite transformations. (5)

4. (a) Explain the following : (10)

   (i) Point clipping

   (ii) line segment clipping

   (iii) convex polygon clipping

   (b) Explain Sutherland Hodgmann algorithm with diagram. (10)

5. (a) Explain the various three dimensional transformation. (15)

(b) Write short notes on 3-D composite transformation. (5)

6. (a) Explain Z- Buffer algorithm. (5)

(b) Explain Cohen Sutherland algorithm for 3-D clipping. (15)

7. (a) Explain the perspective projection of a 3-D objects. (10)

(b) Explain the following :

(i) Mirror reflection.

(ii) Aspect ratio. (10)

8. (a) What are the various components of user Interface? Explain. (10)

(b) Write short notes on Command languages. (10)

DE–3603 **31**

DISTANCE EDUCATION

B.C.A / B.C.A (Lateral) DEGREE EXAMINATION, MAY 2012.

BUSINESS COMMUNICATION

(2003 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) What is communication? Explain the various forms of communication. (10)

   (b) Describe the important goals of communication. (10)

2. Explain the various parts of business letters. (20)

3. (a) Draft a letter placing order for computer systems. (10)

   (b) Is the customer always right? Discuss this with reference to complaints and their settlements. (10)

4. (a) Explain the importance of Minutes Writing. (10)

   (b) Describe the various essential elements of Business Report. (10)

5. (a) What is an Essay? Give the structure of an essay. (10)

(b) What do you mean by presentation? How is it differs from speech. (10)

6. Explain the important matters to be included in the Application for a situation. (20)

7. (a) What are the various types of Testimonials? (10)

(b) Draft an Application for the post of Computer Programmer. (10)

8. (a) Write an essay on scope of employment opportunities for B.C.A. students. (10)

(b) Give a specimen of a Director Report. (10)

DE–3604 **32**

DISTANCE EDUCATION

B.C.A./B.C.A. (Lateral) DEGREE EXAMINATION, MAY 2012.

WINDOWS AND VISUAL BASIC

(2003 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

(5 × 20 = 100)

1. (a) Explain the basic concepts of Windows Programming. (10)

   (b) Describe the menus and its components. (10)

2. (a) (i) What is multitasking? Explain. (5)

   (ii) Write short notes in text formatting. (5)

   (b) Explain the picture attributes. (10)

3. (a) What is OLE? Explain. (10)

   (b) Explain the following.

   (i) Multimedia Windows (5)

   (ii) GUI. (5)

4. (a) How do you use controls and events in VB? Explain. (10)

   (b) Discuss the following.

   (i) Frames (5)

   (ii) Menus. (5)

5. (a) How do you write, save and run a VB code? Explain. (10)

(b) Write a VB program to check whether a string is palindrome or not. (10)

6. (a) Explain the FOR... NEXT statement with example. (10)

(b) Describe the Following with example.

(i) While ... Wend statement (5)

(ii) CASE statement. (5)

7. (a) Write a VB program to find the Factorial of a number using recursion. (10)

(b) How do you create procedures and functions in VB? (10)

8. (a) Discuss the following.

(i) Arrays and control arrays (5)

(ii) MDI forms. (5)

(b) Explain how do you use event handling and indexing. (10)

DE–3605 **33**

DISTANCE EDUCATION

B.C.A/B.C.A.(Lateral) DEGREE EXAMINATION, MAY 2012.

INTERNET AND JAVA PROGRAMMING

(2003 Onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

Each question carries 20 marks.

$(5 \times 20 = 100)$

1. (a) Discuss about Network Topologies. (10)

   (b) Differentiate between TCP and IP. (10)

2. (a) Give short note on Mail Transfer Protocol and File Transfer Protocols. (10)

   (b) How do you create the Web Server? Explain with an example. (10)

3. Discuss elaborately the use of Internet at school. (20)

4. (a) Explain in detail about HTTP and HTTP client requests. (10)

   (b) Write about Fragmentation and routing. (10)

5. (a) Write short notes on Java datatypes. (10)

   (b) Explain arrays in Java. (10)

6. (a) List out the Java operators. Describe the arithmetic operators with examples. (10)

(b) Describe Type Conversion and casting with suitable examples. (10)

7. (a) Discuss about creating objects and accessing class members with an example. (10)

(b) Explain Inheritance with an example. (10)

8. (a) Write a program to extract a portion of a character string and print the extracted string. Assume that m characters are extracted, starting with the nth character. (10)

(b) What is an applet? Discuss the steps involved in developing and running a local remote applet. (10)

DE–3606 **34**

DISTANCE EDUCATION

B.C.A/B.C.A (Lateral) DEGREE EXAMINATION, MAY 2012.

RDBMS

(2003 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks. ($5 \times 20 = 100$)

1. (a) Describe about different user interfaces. (10)

   (b) Explain about schema and subschema in detail. (10)

2. (a) Explain the concepts of Relational Model. (10)

   (b) Write about mapping operation. Give an example. (10)

3. (a) Discuss the benefite of oracle. (10)

   (b) Explain the Logical structures in oracle with example. (10)

4. (a) Discuss about DML. (10)

   (b) Discuss on the basic queries using expressions. Show that with examples. (10)

5. (a) Write SQL query to create an employee table with the fields emp no, name, designation, basic pay, DA, HRA. (12)

   (b) Write SQL query to calculate the gross pay and net pay and display the employee details. (8)

6. Explain on how to create a table, modify a table, Rename a table, copying from another table and to drop the table with examples. (20)

7. (a) Write in about client/server system. (10)

   (b) What is Database schema? Explain. (10)

8. (a) Write short notes on Table definitions. (10)

   (b) Explain the elements of SQL Language. (10)

DE–3607 **35**

DISTANCE EDUCATION

B.C.A./B.C.A. (Lateral) DEGREE EXAMINATION, MAY 2012.

MANAGEMENT PRINCIPLES AND TECHNIQUES

(2003 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

$(5 \times 20 = 100)$

All questions carry equal marks.

1. Explain the various management functions.

2. List out and explain the various theories of motivation in management.

3. The following table gives optimistic time (a), the most likely time (m), and the pessimistic time (b).

Activity

	1-2	1-3	2-5	3-4	4-5	5-8	4-6	4-7	6-9	8-9	7-10	9-10
a	3	1	6	8	0	5	6	3	1	3	8	2
m	5	2	8	12	0	7	9	6	2	5	15	5
b	7	3	12	17	0	9	2	8	3	8	20	6

(a) Draw the network of the project.

(b) Calculate the free float and total float for each event.

4. Solve graphically the following linear programming problem.

Min. $z = 3x_1 + 5x_2$

Subject to :

$-3x_1 + 4x_2 \leq 12$

$2x_1 - x_2 \geq -2$

$2x_1 + 3x_2 \geq 12$

$x_1 \leq 4,\ x_2 \geq 2$

$x_1, x_2 \leq 0.$

5. Find the optional solution for the cost and supply/demand matrix as given below :

	S_1	S_2	S_3	S_4	Supply
O_1	1	2	1	4	30
O_2	3	3	2	1	50
O_3	4	2	5	9	20
Demand	20	40	30	10	100

6. A machine owner funds from his past records that the cost per year of maintaining a machine A whose purchase price is Rs. 6,000 are as given below :

Year :	1	2	3	4
Maintenance cost Rs. :	1,000	1,200	1,400	1,800
Re-sale price Rs. :	3,000	1,500	750	375
Year :	5	6	7	8
Maintenance cost Rs. :	2,300	2,800	3,400	4,000
Re-sale price Rs. :	200	200	200	200

As what age is a replacement due?

7. A research team is planned to rise to a strength of 50 chemists and then to remain at that level. The wastage of recruits depends on their length of service which is as follows.

Year :	1	2	3	4	5
Total percentage who have left upon the end of the year :	5	36	56	63	68
Year :	6	7	8	9	10
Total percentage who have left upon the end of the year :	73	79	87	97	100

What is the recruitment per year necessary to maintain the required strength? There are 8 senior posts for which length of service is the main criterion. What is the average length of service after which new entrant expect his promotion to one of these posts?

8. Solve minimal assignment problem whose effectiveness matrix is

	I	II	III	IV
A	2	3	4	5
B	4	5	6	7
C	7	8	9	8
D	3	5	8	4

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DISTANCE EDUCATION

B.C.A./B.C.A. (Lateral) DEGREE EXAMINATION, MAY 2012.

LAB II — C++ AND MS OFFICE

(2003 onwards)

Time : Three hours Maximum : 100 marks

Examiner has to select and give ONE question to each candidate by Lot system.

1. (a) Write a C++ program to read and display the students particulars such as reg. no., name, age and sex using class and objects.

   (b) Create a library table with the following fields :

   Book No., Title, Author, Price, Publisher, Branch.

   (i) Add 5 records using FORMS.

   (ii) Display the records for given title and author.

   (iii) Display the records for price between 300 to 500.

   (iv) Sort the table in ascending order of book no.

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2. (a) Write a C++ program to find the square of a given number with different arguments using function overloading.

   (b) Create the following worksheet in Excel :

	A	B	C	D	E
1		Amount in Crores			
2	Year	Sales	Expenses		
3	1990	20	9		
4	1991	25	12		
⋮	⋮	⋮	⋮		

   Draw the graph and mention the appropriate headings.

3. (a) Write a C++ program to illustrate such as addition, subtraction, multiplication and division using a member function which are defined out of the scope of a class definition.

(b) Using MS-Word type out the title page of your office automation book.

Format the title page by the following formatting features.

(i) Centre each line

(ii) Choose font type where needed

(iii) Use color options as you feel appropriate.

---------- cut here ----------

4. (a) Write a C++ Program to read the derived class data members such as name, roll no., sex, height and weight from the keyboard and display the contents of the class on the screen. (Single inheritance concept).

(b) Using the MS-word type a circular letter to be posted to PGDCA candidates to attend PCP. Format the letter and apply the suitable font type and size.

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5. (a) Write a C++ program to create a class of objects e_1, e_2. The conents of object e_1 is added to the object e_2 using the operator overloading technique.

(b) Create the worksheet with the following fields :

Reg. No., Name, marks for 3 subjects and total.

(i) find the total column

(ii) filter the candidate whose name = 'Ramu'

(iii) filter the candidate whose total > 240.

6. (a) Write a C++ program to sort the given set of numbers in ascending order.

(b) Create Address table with the following fields :

name, age, sex, street, city, pincode.

(i) add five records

(ii) display the fields name and street alone on the screen

(iii) display the records for age > 20 and city = 'TRICHY '

(iv) sort the tables in the ascending order of name.

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7. (a) Create a class called employee that contains employee no., employee name, designation, basic pay, deductions (LIC, PF) Include member function to get data from user for 'n' employees. Write a C++ program to prepare the payslips for 'n' number of employees.

HRA = 450 , DA = 40% of basic pay

Gross pay = Basic pay + DA + HRA

Net pay = Gross pay - deductions.

(b) Create the worksheet with the following fields :

S.No. Reg. No., Name, Age, Maths, English, Science, Total, Average, Result.

(i) Fill the total column.

(ii) Fill the average column.

(iii) Replace the result with 'PASS' per average > = 50 else 'FAIL'.

(iv) Sort this list by name then age.

8. (a) Write a C++ program to find the sum and average of given set of marks.

(b) Create sales table with the following fields :

name, sex, sales code, sales amount

(i) add 5 records using FORMS

(ii) display the records for sales amount > 50000 and sex = 'M'

(iii) display the records for sex = 'F' and sales amount between 10000 and 20000.

(iv) sort the table in ascending order of sales amount.

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DISTANCE EDUCATION

B.C.A./B.C.A. (Lateral) DEGREE EXAMINATION, MAY 2012.

LAB III — JAVA AND VISUAL BASIC

(2003 onwards)

Time : Three hours Maximum : 100 marks

Examiner has to select TWO questions. One from Java and another from visual basic for each candidate by lot system.

Each question carries 50 marks.

JAVA

1. (a) Write a java program to convert a given number into words for numbers 1 to 10. (25)

   (b) Write a java program to calculate the area of rectangle and Right-angled triangle. (25)

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2. Write a java program to compute commission earned by a salesman according to the scheme given below:

SALES AMOUNT	COMMISSION %
UPTO Rs.1000	0
ABOVE 1000 UPTO 5000	5
ABOVE 5000	10

3. There are 10 students in a class. Their names and marks in three different subjects are given. If a student takes more than 40 marks in each subject, then he is declared 'PASS'. Otherwise 'FAIL'. Write a java program to do the above.

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4. Write a java program that receives the data such as age and name of person to check the eligibility for voting. Take the condition that if a person is more than 18 years old he is eligible to vote. Else display the number of years, he has to wait for voting.

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5. (a) Write a java program to find the biggest of 5 numbers. (25)

   (b) Write a java program to display the prime numbers between 1 to 100. (25)

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VISUAL BASIC

6. Write a VB program to compute the surface area and volume of a sphere given the radius. Use option buttons and Input Boxes.

---------- Cut here ----------

7. Write and test a VB program to compute and print either the SUM or the PRODUCT of the first N natural numbers. Use option Button.

8. Write a VB program to do temperature conversion C to F and F to C at user's choice using label, text and Enter key.

---------- Cut here ----------

9. Write and test a VB program that allows the user choice among four arithmetic operations of addition, subtraction, multiplication and division with two given numbers.

---------- Cut here ----------

10. Write and test a VB program to select candidates for four posts, based on their qualifications.

MALE AND PG DEGREE	MANAGER
MALE, GRADUATE AND TYPING	CLERK
FEMALE AND PG DEGREE	SECRETARY
FEMAL, GRADUATE AND TYPING	STENO

DE–4355 **36**

DISTANCE EDUCATION

B.C.A./ B.C.A. (Lateral) DEGREE EXAMINATION, MAY 2012.

LAB III — ORACLE AND VISUAL BASIC

(Common for all Batches)

(2003 onwards)

Time : Three hours Maximum : 100 marks

Examiner has to select and give ONE question to each candidate by lot system.

1. (a) Write a VB program to compute and print either sum or the product of first N natural numbers. Use option button.

(b) Create table MARK with the following fields.

REGNO, NAME, TAMIL, ENGLISH, MATHS, BIOLOGY

(i) Add 5 records.

(ii) Show data in fields REGNO, NAME, TAMIL and BIOLOGY.

(iii) Show data in fields NAME and REGNO alone on the screen.

(iv) Show all fields on the screen.

2. (a) Write a VB program to do temperature conversion C to F and F to C at user's choice using label, text and enter key.

(b) Create table ADDRESS with the following structure.

NAME	Character	20
STREET	Character	20
CITY	Character	20
PIN	Numeric	6

(i) Add 5 records.

(ii) Display the structure of the table.

(iii) Add the field DIST

(iv) Fill all DIST with TRICHY.

---------- Cut here ----------

3. (a) Write a VB program to read in principal, number of years and rate of interest through INPUT boxes, compute and print the simple interest and compound interest through labels.

(b) Create table LABEL with the following fields.

NAME, STREET, CITY, PIN

(i) Add 5 records.

(ii) Display the contents of table

(iii) Display the descending order sorted list with CITY as primary key.

4. (a) Write a VB program to select candidates for four posts, based on their qualifications.

Male and PG Degree	Manager
Male, Graduate and Typing	Clerk
Female and PG Degree	Secretary
Female, Graduate and Typing	Steno

(b) Create table MARKS with the fields.

REGNO, NAME, MARK 1, MARK 2, MARK 3, TOTAL

(i) Add 5 records.

(ii) Replace all TOTAL with MARK 1 + MARK 2 + MARK 3

(iii) Display the maximum marks in each subject.

(iv) Count the number of records for MARK 3 > 80.

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5. (a) Write a VB program to calculate simple interest or compound interest using three command buttons namely input, calculate and display.

(b) Create table BLOOD with the following fields.

DNO, DNAME, DOB, DAGE, DADD, DBLOOD, DSEX

(i) Add 5 records.

(ii) Display all records.

(iii) Display the information of donors having date of birth between 01 – JAN – 95 and 31 – DEC – 2005.

6. (a) Write a VB program to display the day of the given date.

(b) Create table RESULT with the following fields.

REGNO, NAME, MARK 1, MARK 2, MARK 3, TOTAL, RES

(i) Add 5 records.

(ii) Fill TOTAL field with the sum of MARK 1, MARK 2 and MARK 3.

(iii) Fill the RES field with 'PASS' if TOTAL > = 120 otherwise 'FAIL'.

---------- Cut here ----------

7. (a) Write a VB program to that allows the user choice among four arithmetic operations of addition, subtraction, multiplication and division with two given numbers.

(b) Create table PERSON with the following fields.

SNO, NAME, CITY, PIN, PHONENO

(i) Add 5 records.

(ii) Display all names.

(iii) Display all names without duplication.

(iv) Display name, city in which only beginning portion of name is known.

---------- Cut here ----------

8. (a) Write a VB program to compute the surface area and volume of a sphere given the radius. Use option button and input boxes.

(b) Create table SALES with the following fields.

SALENO, SALENAME, SALEAMT, COMM, SALEDATE

(i) Add 5 records.

(ii) Compute COMM as per your logic

(iii) Display all records.

(iv) Display the average sales and average commission.